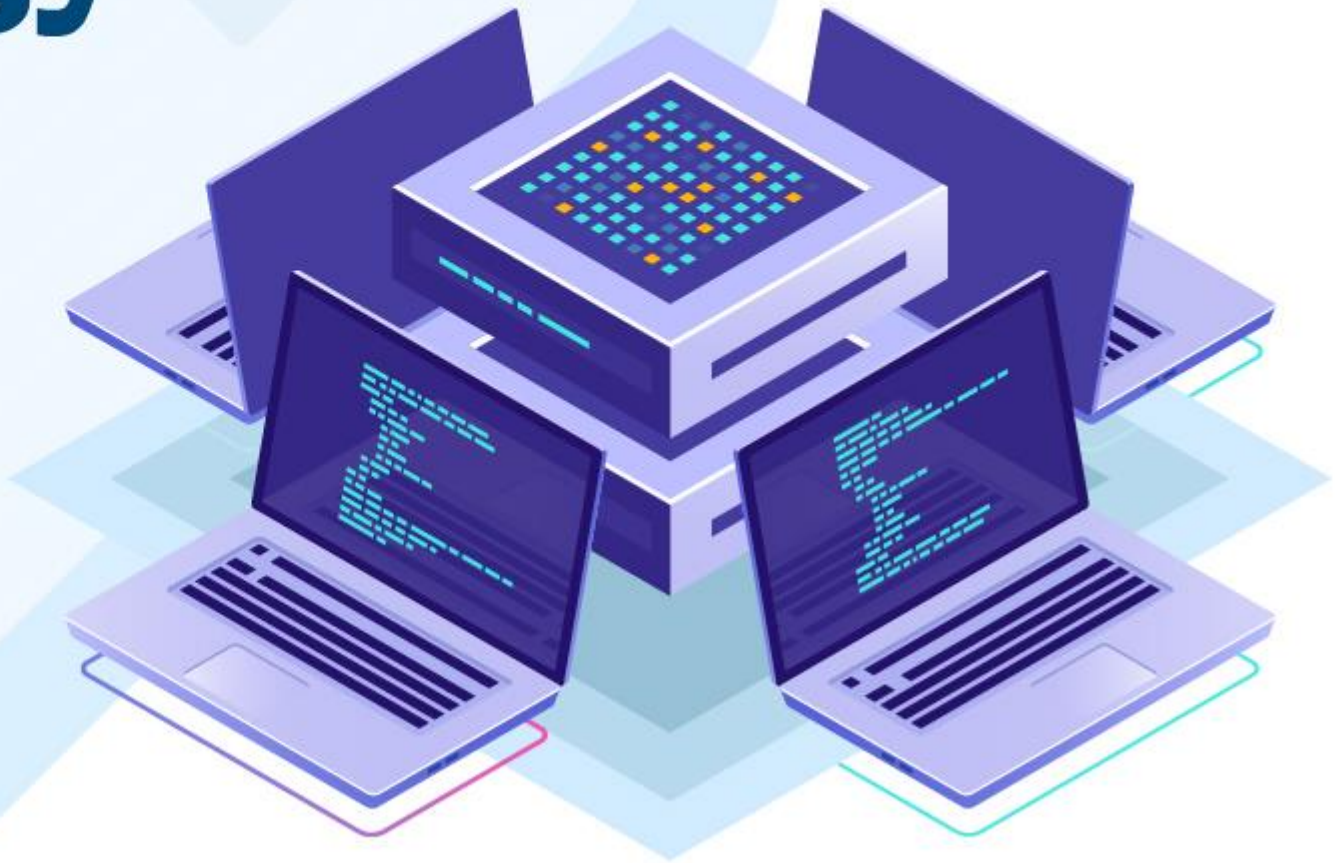
Information Technology

Fall Semester
2026 - 2027

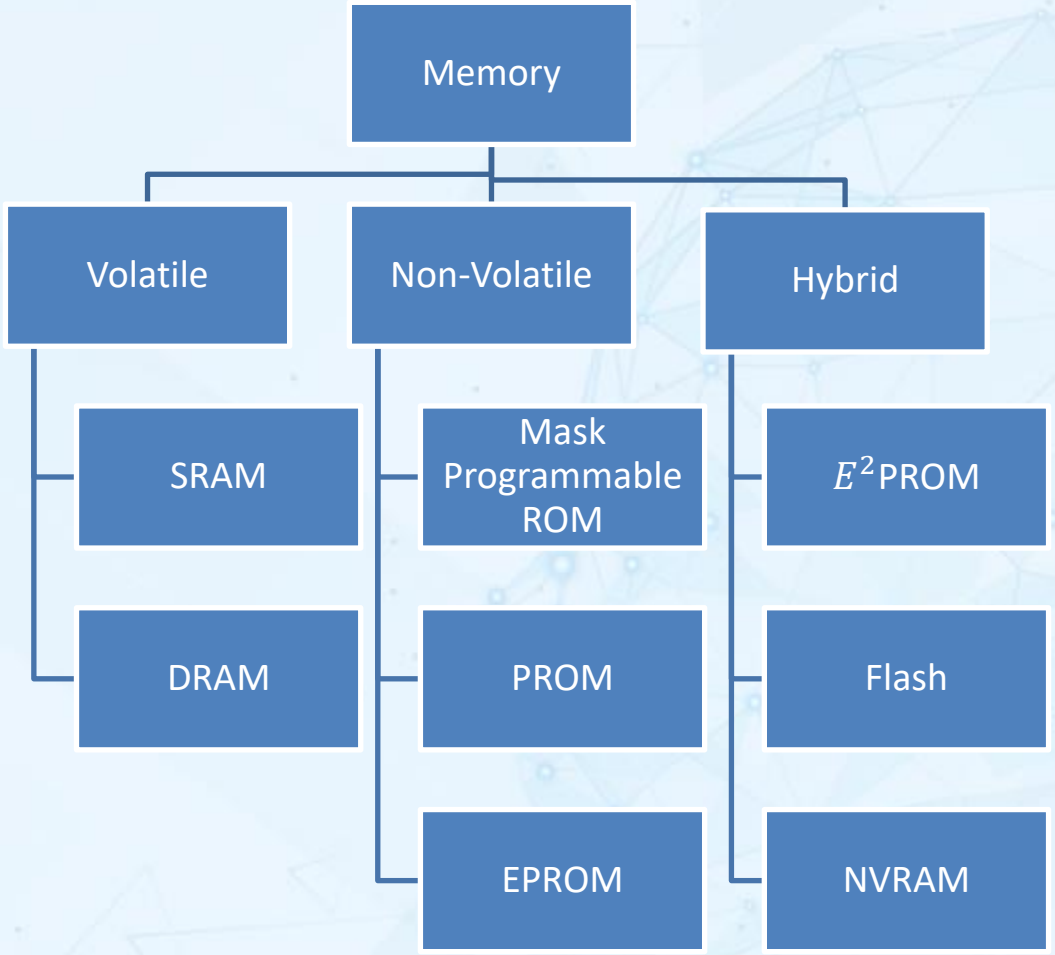
- Embedded Systems
- **Dr. Yasmine Mahmoud**



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Memory Types

1. Three types of memory:



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Memory Types

1. Three types of memory:
   - Volatile (RAM)
   - Non-Volatile (ROM)
   - Hybrid (Mix between RAM and ROM)

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Memory Types

1. Volatile (RAM)
   - The data still there if the power is connected.
   - It doesn’t keep data if there is no power.
2. Non-Volatile (ROM)
   1. The data still there if the power connected or disconnected.
3. Hybrid

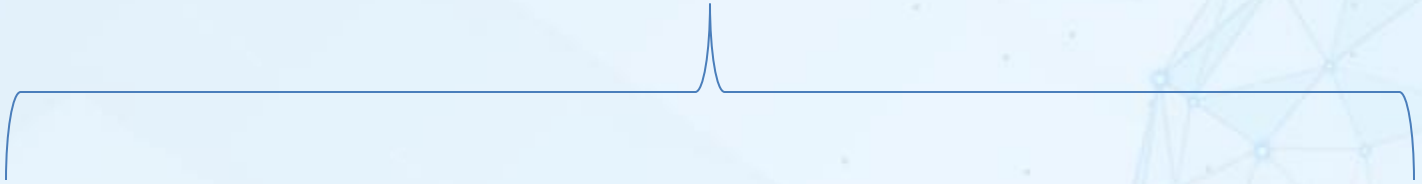
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Volatile Memory (RAM) Types

- Random Access Memory (RAM) or Read & Write Memory (RAM).
- Random Access Memory : it is fast to any memory location.
- RAM is faster than ROM.
- RWM (working memory)
- Why?
- Because of the components the RAM and ROM consists of. As we will see later.

Volatile Memory (RAM) Types

- There are two types of RAM:

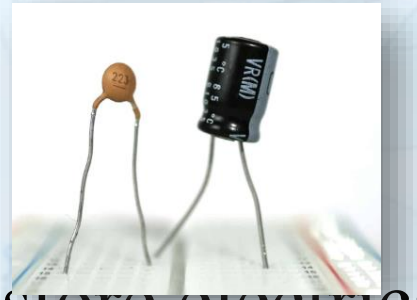


SRAM	DRAM
Static RAM	Dynamic RAM

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Volatile Memory (RAM) Types

1. Dynamic RAM (DRAM)



- Based on capacitor
- Capacitors are electronic components that can store electrical charge.
- It store an electrical charge, to represent a 1, or no charge, to represent a 0
- If the capacitor is charged it will hold value. “bit represents 1”
- If the capacitor is discharged, it will hold another value. “bit represents 0”

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Volatile Memory (RAM) Types

1. Dynamic RAM (DRAM)

- What is the problem of using capacitor?
  - The primary issue with capacitors, particularly older electrolytic capacitors, is their tendency to lose capacitance over time, leading to reduced performance and potential circuit malfunctions.
  - The capacitor discharges over time.
  - If we charge it value = 1.
  - Overtime it will be zero.
  - So, we need a **Refreshment Circuit** to prevent charge leakage.

Volatile Memory (RAM) Types

1. Dynamic RAM (DRAM)

– **Refreshment Circuit (DRAM Controller):** it is a circuit used to periodically read and write back the data stored in DRAM cells. So, it access the memory.

– We also have problem now !!!

– **That both Microprocessor (CPU) and Refreshment Circuit access the memory.**

– Who will have the highest priority to access?

- Refreshment Circuit.
- If Refreshment Circuit doesn't refresh the value of 1, it will be zero, so there is no point of memory as the values inside the memory will be lost or corrupted.

Volatile Memory (RAM) Types

1. Dynamic RAM (DRAM)
   - DRAM is slow as it will delay the CPU (DRAM has higher priority to access memory than CPU)
   - So, the access time will increase, and the speed will decrease.

Volatile Memory (RAM) Types

1. Dynamic RAM (DRAM) Advantages
   - DRAM is simple hardware.
   - Low cost per bit. As it has simple hardware “capacitor”.
   - High density. (can buy big size as it is low cost)

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Volatile Memory (RAM) Types

2. Static RAM (SRAM) Advantages

- Based on Transistor
- A bit is represented by a flipflop
- one flipflop At least use 6 transistors
- SRAM doesn't need a refreshment circuit. So, it is faster than DRAM.
- Problem: SRAM is high cost per bit. (consists of 6 transistors for each bit)
- In Embedded Systems , we use SRAM with small size as it is expensive.

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Volatile Memory (RAM) Types

Comparison between SRAM & DRAM

ES Challenges	SRAM	DRAM
Size	Low	High density
Cost	high	low
Performance	faster	slower
Power consumption	Lower	Higher (need refreshment circuit, and the capacitor consume power)

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Non-Volatile (ROM) Types

- Read Only Memory (ROM):
  - It is the program memory, which means that it is the memory that hold the code.
  - Access time of ROM is slower than the access time of RAM.
  - ROM is based on floating gate MOSFET (FGM), which are commonly used in certain types of non-volatile memory like Flash memory.

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THANK YOU FOR WATCHING

QUESTIONS?

